

HK-ClaimSentry

Hybrid C++17 + Python ML Claims Adjudication Engine for Hong Kong Virtual Insurers

Target: Bowtie | ZA | Blue | OneDegree | Author: Nadkalpur Manjunath | github.com/nsmanju/HK-ClaimSentry

Metric	Value	Verification
Flag Rate	3.1% (308 / 10,000 claims)	Matches HK industry baseline
Throughput	500,000 claims / sec (C++ core)	rule_engine.cpp benchmark
Native Bridge	408 KB .so via pybind11	claim_engine_cpp.cpython-312.so
API Latency	< 5 ms p95 for /scrutinize	FastAPI Uvicorn 127.0.0.1:8000

Live API Verification — 2026-09-27

POST /scrutinize HTTP/1.1 200 OK
Host: 127.0.0.1:8000
Content-Type: application/json

Request:

```
{
  "hospital_code": "QEH",
  "diagnosis_en": "common cold",
  "amount_hkd": 3083,
  "hko_signal": 0
}
```

Response:

```
{
  "cpp_flagged": true,
  "reason": "FLAGGED: Public HA QEH cannot bill HKD 3083 for common cold",
  "risk_score": 0.95,
  "final_flag": true
}
```

Build: cmake .. && make — Built target claim_engine_cpp
Runtime: python3 -m uvicorn python_core.api:app --port 8000 — /docs

Rules Engine — Weighted Adjudication

ID	Rule Description	Example Trigger	Weight	System Action
R1	Public Hospital Overbilling	QEH — Common Cold HKD 3,083	0.95	BLOCK — Exceeds HA subsidy cap HKD 800
R2	Typhoon Signal Impossibility	Gleneagles — T9 Signal Present	0.85	CHECK — Transport suspended per HKO
R3	Private Facility Overcharge	Gleneagles — Cold HKD 9,948	0.75	REVIEW — Exceeds private benchmark HKD 1,800
R4	ML Anomaly Detection	Unseen pattern	ML Score	FLAG — IsolationForest contamination 0.03

Architecture

C++17 Core (cpp_core/src/rule_engine.cpp) — Deterministic rule evaluation <2µs per claim.

pybind11 Bridge (pybind_wrapper.cpp → .so) — Zero-copy interop, built via CMake.

Machine Learning (python_core/ml/model.py) — IsolationForest for unknown anomaly detection.

FastAPI Service (python_core/api.py) — /scrutinize endpoint with OpenAPI docs at /docs.

Data Model — HK-specific: QEH/HKSH/Gleneagles, HKO T8/T9/T10 signals, realistic pricing (Public HKD

180-800, Private HKD 400-1800).

Compliance Notes

Public hospital pricing aligned with Hospital Authority (HA) subsidized outpatient cap. Typhoon logic aligned with Hong Kong Observatory (HKO) signal definitions and MTR service suspension policy. Private hospital benchmarks derived from HK private sector fee schedules.

HK-ClaimSentry — Hybrid C++17 + Python ML Claims Engine | GitHub: github.com/nsmanju/HK-ClaimSentry
API: `uvicorn python_core.api:app --reload --port 8000` → <http://localhost:8000/docs> | Contact: Nadkalpur Manjunath